Assignment 5 Report

Project Members

Beckham Reyes
beareyes@ucsc.edu
BeReyes1

Hao Deng
hdeng25@ucsc.edu
ACDragon

Dominic Umbrasas
dumbrasa@ucsc.edu
https://github.com/Dominic-Um

Charles Haiwen
chaiwen@ucsc.edu
https://github.com/Magmyte

Game Expanded

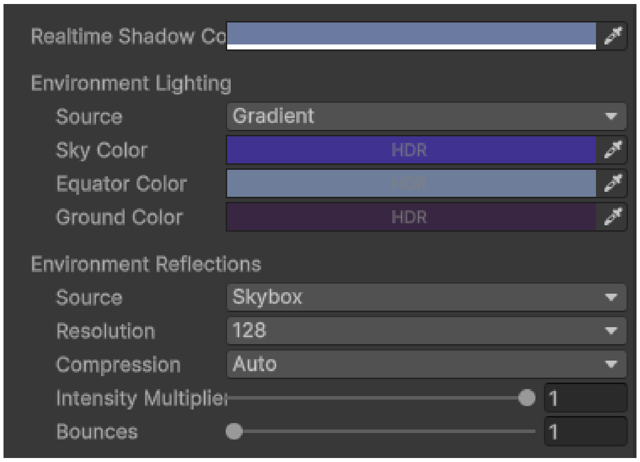
Stealth Game

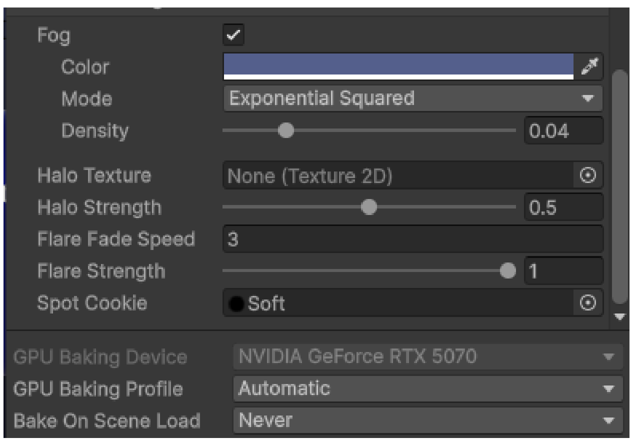
Summary of Additions

-Main Menu, Settings, Credits
-4 new levels with level transitions
-2 mechanics: key, guard disguise
-Custom lighting/lighting variations
-Level assets

Graphics Integration

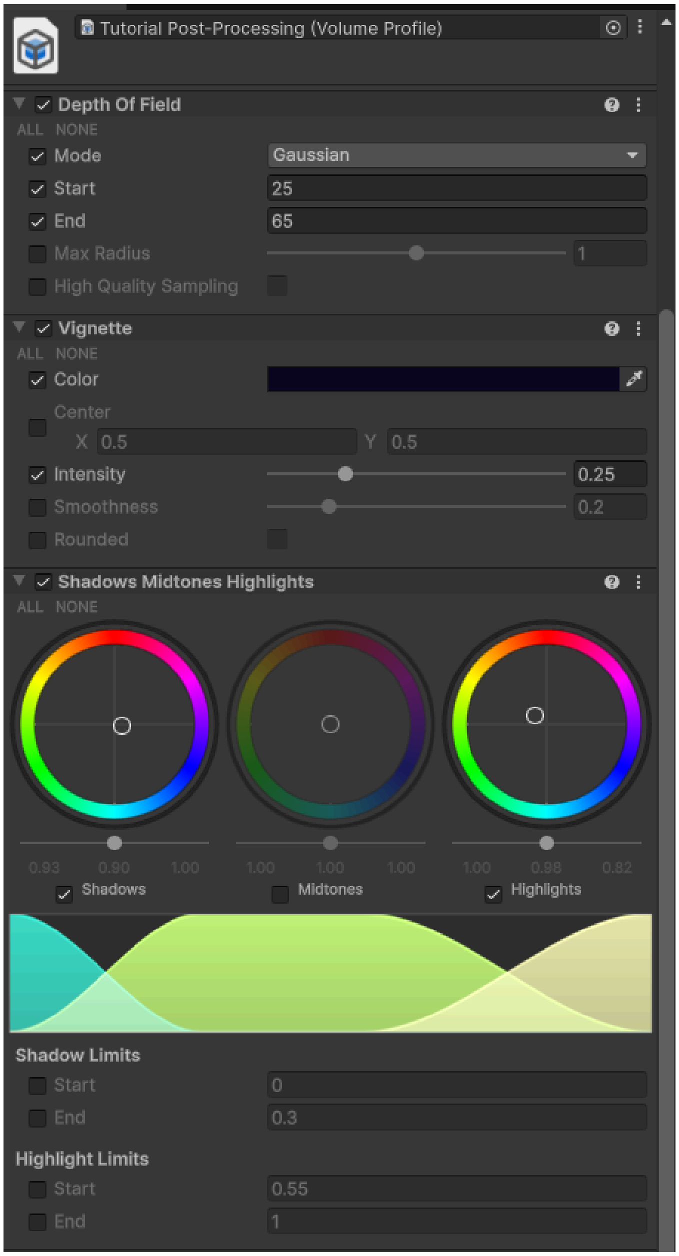
Lighting





We didn’t use a directional light to light up the scene as we instead use the environment tab in the lighting settings to control it. It leans towards a purple color to give the dark coloring of our world. And an important feature was the fog to sell a snowstorm and the darkness of the night.

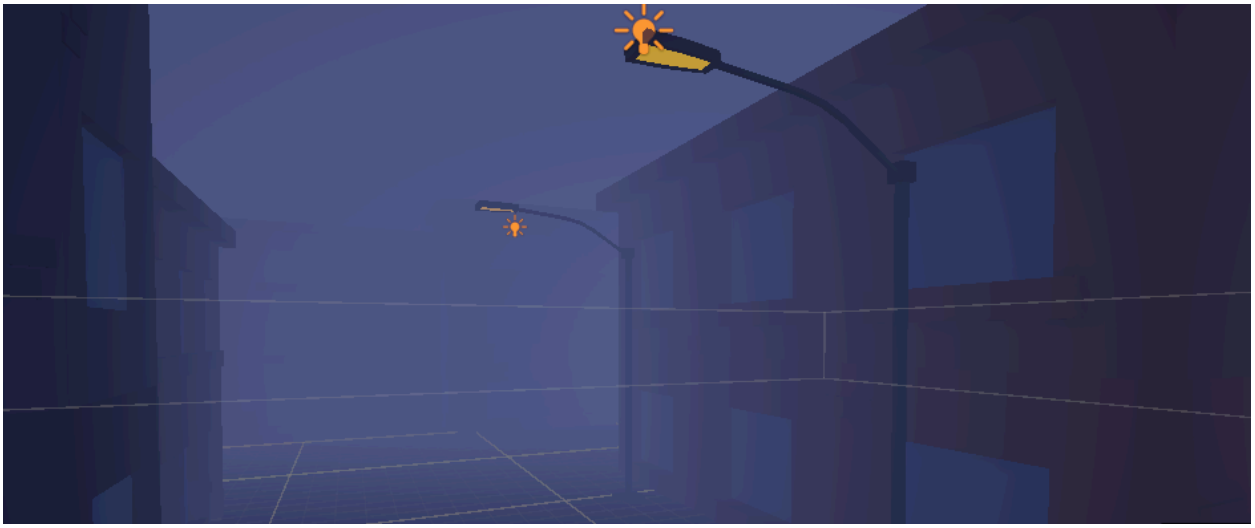
Post-Processing



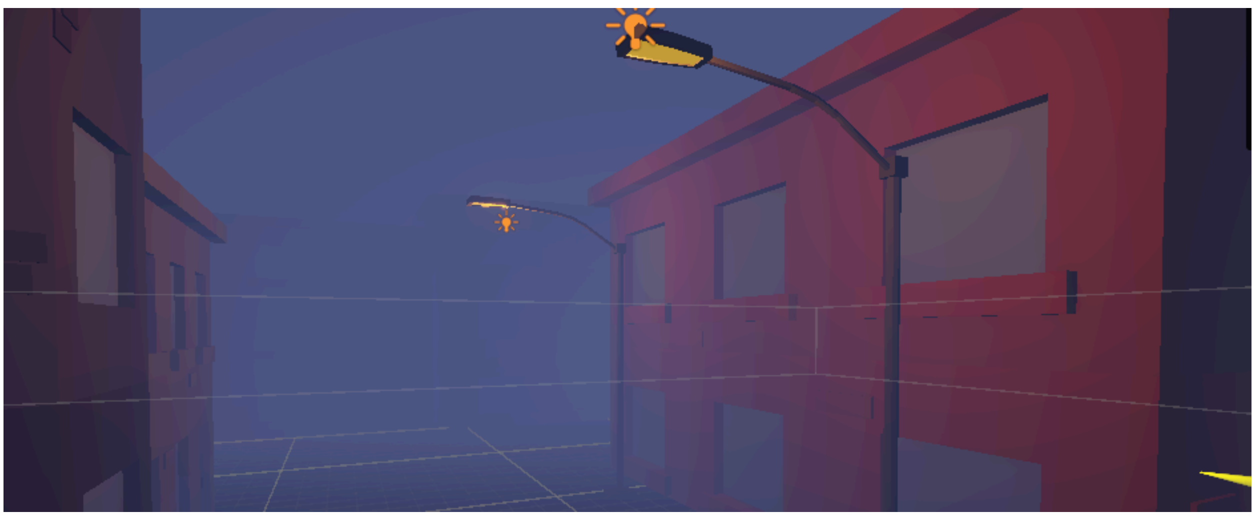
We have three effects on our post processing. The depth of field is subtle as it makes objects in the distance appear slightly blurrier which adds to our fog-like settings. The vignette slightly darkens the corners of the screen to fit the nighttime theme where vision is weakened. But not too much as the player needs to see the guards. Lastly, the shadows and highlights were used to give a stronger purple color to contribute to the darker vibe.

Lighting Variations:

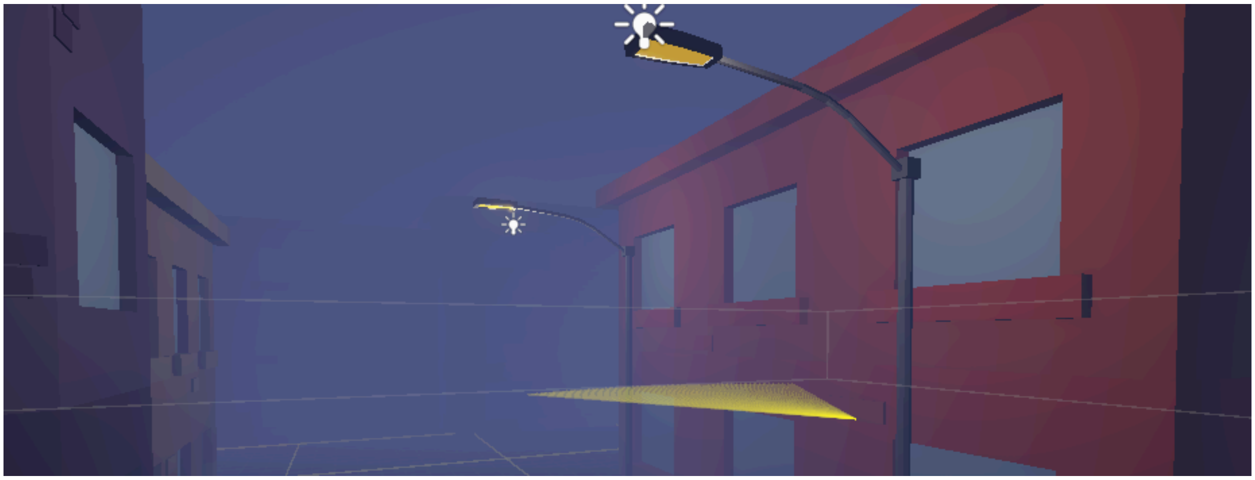
OFF:



Orange Tint:



White Tint:



We have streetlights in our game. The player can manipulate the lighting of them by going up to a lightbox and pressing a button. There is a normal white color, orange color, and the lights completely off. It is purely cosmetic with no effect on the gameplay, although turning the flights off would probably make the game a little bit more difficult due to less vision.

PBR Materials:

All materials in this game are URP/Lit which are PBR materials. Most are metallic while a few are specular.

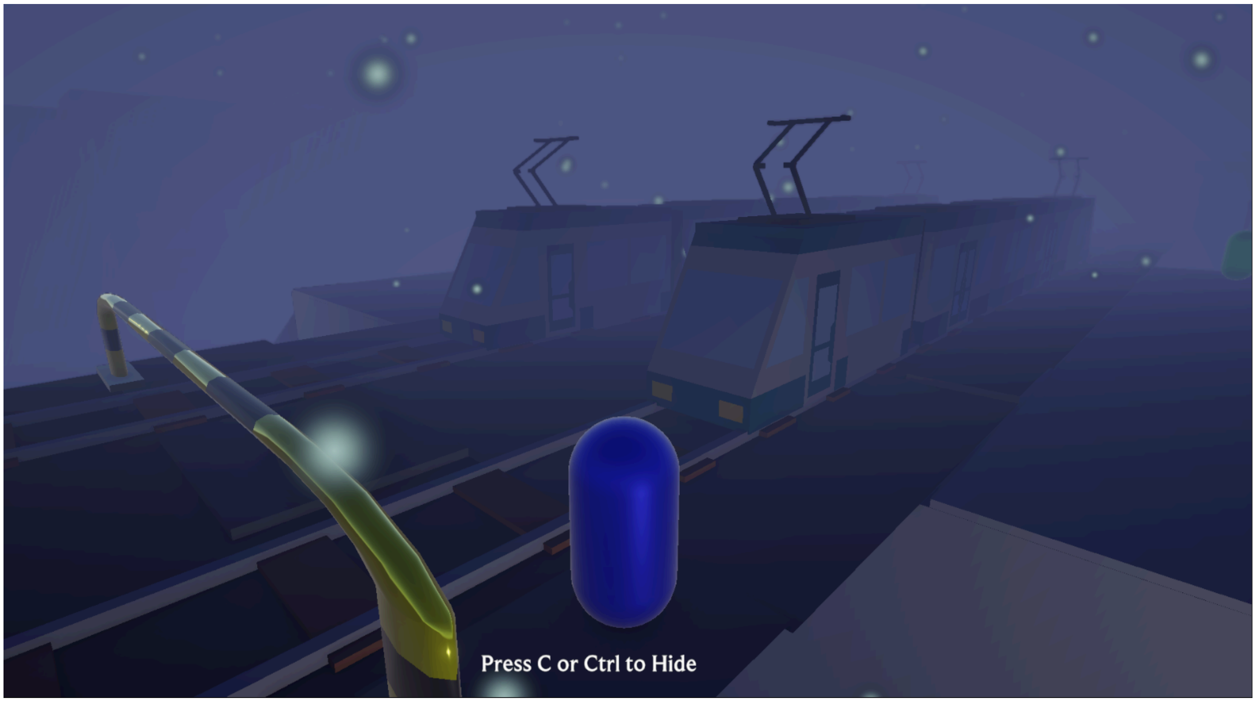
Audio Assets

Listed with attribution

Music:
Cyberpunk Moonlight Sonata by Joth on https://opengameart.org/
Crystal Cave (song18) by cynicmusic on https://opengameart.org/
Free Music Pack - Flags by tricksntraps on https://opengameart.org/

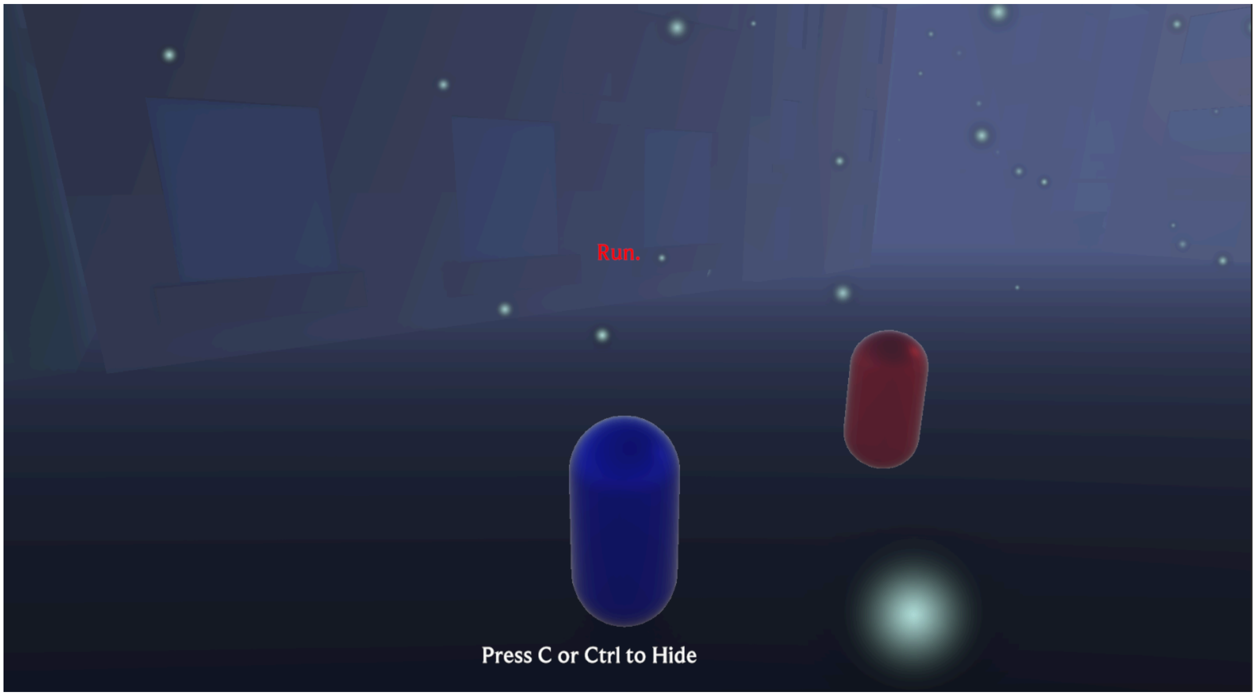
SFX:
50 RPG sound effects - handleCoins by Kenney on https://opengameart.org/
50 RPG sound effects - bookPlace3 by Kenney on https://opengameart.org/
GUI Sound Effects - negative_2 by Lokif on https://opengameart.org/

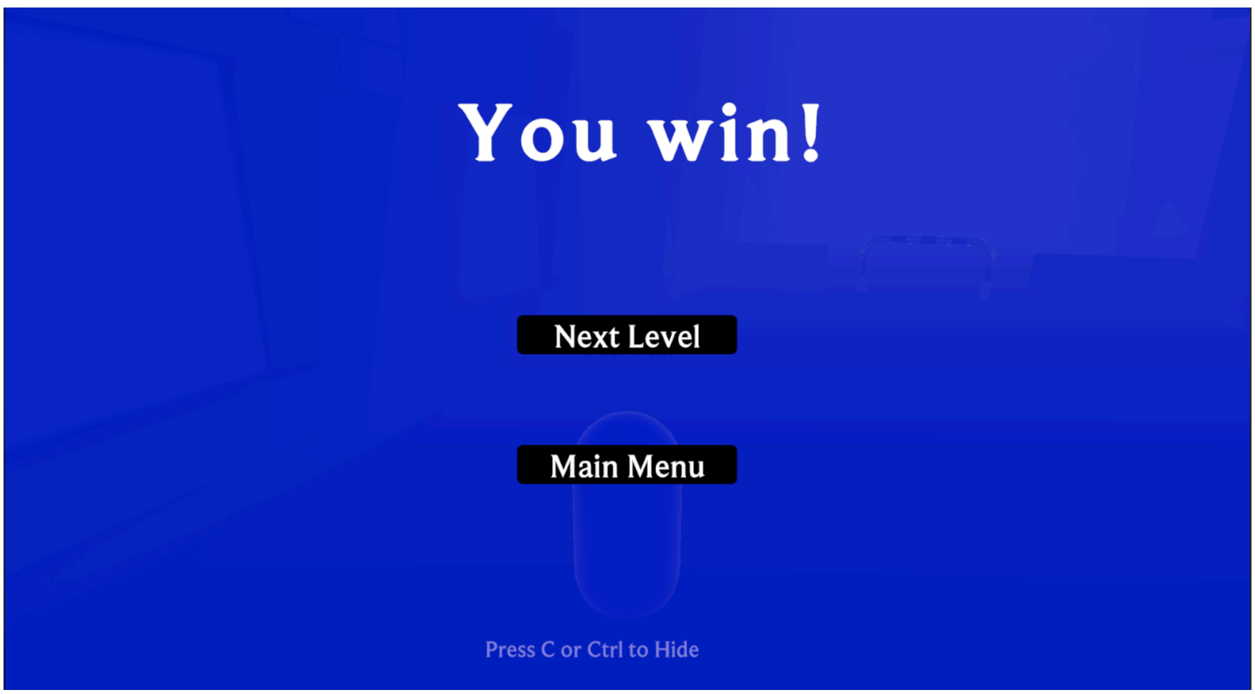
Screenshots




You are now disguised
Press C or Ctrl to Hide


Press E to Toggle Lighting
Press C or Ctrl to Hide


Run.
Press C or Ctrl to Hide


You win!
Next Level
Main Menu
Press C or Ctrl to Hide

Reflection

(2-3) paragraphs

- Your contributions to the project
- Technical challenges you overcame
- What you learned throughout the quarter
- What you would do differently with more time

Beckham Reyes:
I contributed various things. I added UI design and engineering. This includes the screens like the main menu. Also, the ending screen which transitions the player to the next level or the main menu depending on what they choseI I found a font other than the default one for us to use. I also designed all of Level 3 and implemented the guards in the Tutorial. I also did cleanup on a few things like adding additional lights to a level or fixing bugs with enemies.

The technical challenge I had to overcome was handling all of the pull requests as I did almost all of them. That meant I had to go into other people's code or scenes and understand what was going on. Most of the time I understood what was added and how it worked but I occasionally needed to ask a question in our group chat about a feature I didn't understand. There were a few merge conflict issues but they were able to be handled without breaking the project.

I understood most of the concepts from previous experience but I did learn a few things. With lighting I learned how to place reflection probes the right way. I also learned about fog and using the sky color in the environment lighting tab. With more time we could probably improve the design of the levels and some visuals as well.

Hao Deng:

I worked on adding a lighting gameplay mechanic to the game by creating an invisible collider attached to a light and counting the instances of light the player had entered and exited, but which was ultimately cut from the game due to complexity in level design. I also worked on adding music and sound effects to the game, and implemented the NPC patrol routes for levels 1 and 2.

The biggest technical challenge was trying to get audio on every scene without adding audio managers for every scene. Although that solution would have worked, it would be tedious and prone to error. Instead I made a singleton instance of a music manager and an sfx manager to play sounds between scenes and carry music between scenes. I also had an issue with loading audio clips, so I loaded them all in the main menu and carried them throughout each different scene, as the performance impact would be minimal due to the low amount of audio effects and it would remove the need to load it in every scene.

Throughout the quarter, I learned about many useful tools and concepts that are used to make games. I learned various builtins such as physics and navmesh, that would allow me to build various game systems and components, and understanding the concepts of those builtins allows me to look for similarities and use them within other engines such as Godot. . I also gained valuable insight in the process it takes to make a game, involving more than just building the game systems itself. There are many other tasks such as level design, game design, visual design, and so on that must take place, and not just coding.

If I could do things differently, I would first refactor my code to make the systems cleaner and more understandable. I would also modify the detection system and the NPCs to make them chase better, as currently, you can easily juke the guards when they are close and you can also outrun them even in chase. I would also figure out a better solution to loading audio clips, as it would not work for larger games.

Dominic Umbrasas:

I worked primarily on the items and lighting system. I added a door and key items with their relevant scripts as well as a disguise item and its scripts + relevant UI, which stops enemies from aggroing on sight, and also edited the NPC code to work with it. I added a lighting system prefab which is a physical button, empty object and scripts attached to point lights on the level where you can press it to toggle the main light, alternative (warmer) light and just turn it off completely. I also implemented and placed the lights all over a few levels.

The biggest technical challenge is I would say creating the lighting scripts, wiring everything up the correct way with scripts and to not cause errors in the process as well as navigating and coordinating around the work of others, to not cause merge conflicts. There were more minor ones as well such as using my intuition to realize the door needed two box colliders to work properly.

I learned a lot about Unity, especially scripting and lighting as well as creating physical levels with terrain. I also became more proficient with GitHub where I learned to better integrate GitHub Desktop with my projects and use pull requests than just committing to the main branch.

If I could do the project again, I would make better use of the prefab system and integrate it for my features more early. I would also use the time to add more interesting items to the game in general.

Charles Haiwen:
For this assignment, I was in charge of primary level design and visual aesthetics. Most of the visual assets were free prefabs from the Unity asset store, with anything not a prefab being a mesh constructed using Unity's built-in Probuilder package. I also constructed the global lighting and environmental settings, and implemented fog and a particle system, along with a few post-processing effects, to create a more somber environment for the game to take place in, like navigating a city during a blizzard. I also produced the NavMesh surfaces for the guards to navigate on.

The primary challenge I faced was regarding the amount of work required to design the layouts, then putting those layouts into the game using a combination of invisible Probuilder meshes (for collision) and visible prefabs without collision (to show where players can move). This often took significant time to think about player movement and decision points, using Probuilder to put the collision boxes together (using the limited tools Probuilder offers), and then placing, scaling, and aligning the building prefabs from the Unity asset store. Thankfully since I organized all of my game objects well, the following process of creating NavMesh surfaces and using NavMesh modifier components wasn't very difficult, and only took a few iterations to fix minor mistakes when I finally implemented them.

Much of my work was a reiteration of work I had done in previous assignments, but I will say I learned a good amount of workflow procedure in regards to working on repositories with branches and pull requests, two things I did not use frequently before this assignment. If I had more time to work on this assignment, I would have made the levels more detailed, adding more pathways and dead-ends in the level design while incorporating more visual flair through an improved particle system for emulating a blizzard.